Chapter 34
Exotic Options

Five chapters have deferred material to this one. Everything in it trades over the counter, in the market Section 10.10 described; an exotic option is a dealer's product, and that fact shapes how it is priced, hedged, and sold.

Section 15.2 said every payoff whose value depends on S and t alone satisfies the Black–Scholes–Merton equation without modification, and that a path-dependent one adds a state variable. Section 18.6 said vanna and volga matter here. Section 18.13 said that where gamma diverges, dynamic hedging becomes an impossibility problem and a static hedge is the only answer. Section 19.12 said barrier prices depend on how the surface *evolves*, which no static surface specifies. Section 20.4 promised that a barrier priced on a tree whose nodes straddle the barrier converges appallingly. Section 28.11 said the multi-asset family of options is developed here.

The unifying question is which exotics are genuinely harder than vanillas and which merely look it. The answer is not about payoff complexity. It is about whether the payoff can be replicated by instruments the market quotes, and Sections 34.5 and 34.13 give the test.

34.1 WHAT MAKES AN OPTION EXOTIC

Three features separate an exotic from a vanilla, with very different consequences. Two turn on the path rather than the endpoint, so the notation comes first. Throughout, S_t is the underlying and K the strike, and the path statistics are

$$M_T = \max_{0\le t\le T} S_t, \qquad m_T = \min_{0\le t\le T} S_t, \qquad A_T = \frac{1}{T}\int_0^T S_t\,dt$$

the running maximum, the running minimum, and the average.

Path dependence. The payoff depends on the route, not only the destination. A **barrier option** is a vanilla multiplied by an indicator of whether the path has reached a barrier level H:

$$\text{down-and-out call} = (S_T-K)^+\mathbf{1}_{\{m_T>H\}}, \qquad \text{down-and-in call} = (S_T-K)^+\mathbf{1}_{\{m_T\le H\}} \tag{34.1}$$

with the up-variants replacing m_T by M_T. Written this way, in–out parity (knock-in plus knock-out equals the vanilla, equation (34.10) in Section 34.3) stops being a result and becomes an observation: the two indicators sum to one, so the two options sum to the vanilla.

A **lookback option** replaces the strike or the terminal price by an extreme of the path:

$$\begin{aligned} \text{floating call} &= S_T - m_T, & \text{floating put} &= M_T - S_T,\\ \text{fixed call} &= (M_T-K)^+, & \text{fixed put} &= (K-m_T)^+ \end{aligned} \tag{34.2}$$

The floating pair can never finish out of the money, which is most of the reason they cost what they cost. An **Asian option** replaces the terminal price by the average A_T in one of two places:

$$\text{average price} = (A_T-K)^+, \qquad \text{average strike} = (S_T-A_T)^+ \tag{34.3}$$

and a **cliquet** sums period returns $r_i = S_{t_i}/S_{t_{i-1}} - 1$ over reset dates $0 = t_0 < t_1 < \cdots < t_n = T$, each floored at F and capped at C:

$$\text{cliquet} = \sum_{i=1}^{n} \min\big(\max(r_i,\ F),\ C\big) \tag{34.4}$$

The cap is inside the sum rather than outside it, and Section 34.12 is about what that does to the sign of the volatility exposure.

This is where the model risk lives, because a model calibrated to terminal distributions does not pin down paths. Five sections of this chapter go to the barrier alone for that reason (Sections 34.3 through 34.7), and Section 34.9 takes the lookback and the Asian together.

Multiple underlyings. **Basket options** pay on a weighted sum of several assets, **rainbow options** on the best or the worst of them, and **exchange options** on one asset against another; four payoffs matter:

$$\begin{aligned} \text{basket} &= \Big(\sum_i w_i S_i(T) - K\Big)^+, & \text{exchange} &= \big(S_1(T) - S_2(T)\big)^+, \\ \text{worst-of put} &= \big(K - \min_i S_i(T)\big)^+, & \text{best-of call} &= \big(\max_i S_i(T) - K\big)^+ \end{aligned} \tag{34.5}$$

Here $S_i(T)$ is the price of the ith asset at expiration and w_i the number of units of it the basket holds; the basket of Example 34.8 is $(S_1 + S_2)/2$, so $w_1 = w_2 = 1/2$. The new input is correlation, which Section 19.7 established is not observable in the way volatility is. It enters every one of these payoffs and is quoted in none of them, which is Section 34.10's subject and the reason the worst-of put in Section 34.11 behaves as it does.

Non-standard exercise or payoff. Bermudans, compounds, choosers, and the digital pair (options paying either a fixed cash amount Q or the asset itself if they finish in the money), the last of these carrying the identity that ties it to Black–Scholes:

$$\begin{aligned} \text{cash-or-nothing} &= Q\,\mathbf{1}_{\{S_T>K\}}, & \text{asset-or-nothing} &= S_T\,\mathbf{1}_{\{S_T>K\}}, \\ (S_T - K)^+ &= S_T\,\mathbf{1}_{\{S_T>K\}} - K\,\mathbf{1}_{\{S_T>K\}} \end{aligned} \tag{34.6}$$

A vanilla call is one asset-or-nothing less K cash-or-nothings, which is Section 28.10's reading of $N(d_1)$ and $N(d_2)$ written as a payoff rather than as a price. These are usually the easy ones, because a single-asset payoff at a single date is a function of the terminal distribution, which the vanilla market already reveals.

The taxonomy sorts the chapter: the third group is priced almost model-independently, the first two are not. A desk that treats "exotic" as one category prices some trades far too carefully and others not carefully enough.

34.2 OPTIONS THAT ARE BARELY EXOTIC

Several instruments carry exotic names and almost no model risk.

Packages are combinations of vanillas (Chapter 12's spreads, straddles, and collars) sold as one contract and priced by adding up the pieces.

Non-standard American options, exercisable only in windows or with a strike that steps, need Chapter 20's lattice but no new theory. A **gap option** pays $S_T - K_1$ if $S_T > K_2$ and nothing otherwise, which is a vanilla struck at K_2 plus $K_2 - K_1$ in a digital, so it is priced exactly and inherits the digital's hedging problem of Section 34.8. A **shout option** lets the holder lock in the intrinsic value at one chosen date, and pays at expiration the greater of that locked value and the ordinary payoff: a cliquet whose single reset the holder chooses.

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Compound options, an option on an option, have closed forms under Black–Scholes and are genuinely sensitive to volatility in a way vanillas are not: a compound option is a bet on volatility twice over, since both the outer and inner values depend on it. Their vega is correspondingly large relative to their price. Geske's formula for a call on a call (the right, on payment of K_1 at t_1, to acquire a call struck at K_2 expiring at T) is

$$c_{\text{call}} = S_0 e^{-qT} M(a_1, b_1; \rho) - K_2 e^{-rT} M(a_2, b_2; \rho) - K_1 e^{-rt_1} N(a_2), \qquad \rho = \sqrt{t_1/T} \quad (34.7)$$

where $M(\cdot, \cdot; \rho)$ is the bivariate standard normal distribution function,

$$a_1 = \frac{\ln(S_0/S^*) + (r - q + \sigma^2/2)t_1}{\sigma\sqrt{t_1}}, \qquad b_1 = \frac{\ln(S_0/K_2) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{t_1}$, $b_2 = b_1 - \sigma\sqrt{T}$, and S^* is the asset price at t_1 at which the inner call is worth exactly K_1. That critical price is the only part of the formula requiring numerical work, and it is a one-dimensional root search on Black–Scholes. The correlation entering M is $\sqrt{t_1/T}$ because one Brownian motion carries the price through both dates, which is the whole reason a compound option is a two-dimensional integral rather than a one-dimensional one. The other three differ only in signs. Writing p_{call} for a put on a call, and c_{put} and p_{put} for a call and a put on a put,

$$\begin{aligned} p_{\text{call}} &= K_2 e^{-rT} M(-a_2, b_2; -\rho) - S_0 e^{-qT} M(-a_1, b_1; -\rho) + K_1 e^{-rt_1} N(-a_2) \\ c_{\text{put}} &= K_2 e^{-rT} M(-a_2, -b_2; \rho) - S_0 e^{-qT} M(-a_1, -b_1; \rho) - K_1 e^{-rt_1} N(-a_2) \\ p_{\text{put}} &= S_0 e^{-qT} M(a_1, -b_1; -\rho) - K_2 e^{-rT} M(a_2, -b_2; -\rho) + K_1 e^{-rt_1} N(a_2) \end{aligned}$$

Two things change with the inner option rather than the outer one. In the last two, S^* is the price at which the inner put is worth K_1, not the inner call, so it is a different root search; and a put loses value as the asset rises, so the outer option is exercised below S^* for a call on a put and above it for a put on a put.

Only two of the four are independent. Holding a call on a call and writing a put on the same call delivers that call at t_1 against K_1 paid, whatever the asset does in between, so

$$c_{\text{call}} - p_{\text{call}} = c(S_0, K_2, T) - K_1 e^{-rt_1}, \qquad c_{\text{put}} - p_{\text{put}} = p(S_0, K_2, T) - K_1 e^{-rt_1}$$

and those two identities are the cheapest check available on an implementation of any of the four.

Figure 34.1 puts the volatility claim in numbers. At a 10% volatility the compound is worth 3.02 against the vanilla's 6.81, and its value moves **2.2 times** as much for a given relative move in volatility; by 50% that ratio is down to 1.2. The ratio is one of volatility elasticities, $\sigma\,(\partial V/\partial\sigma)/V$, the percentage change in value for a one percent change in volatility. The compound's barely moves: it sits near 1.1 throughout, so its value is close to proportional to volatility, while the vanilla's climbs from 0.50 to 0.86. Paying twice for volatility does not buy a larger vega; it buys a vega that stays large relative to a much smaller price. A compound option suits a firm bidding for a foreign-currency contract it may not win: it fixes the cost of the hedge if the bid succeeds and costs only the first premium if it fails.

Adding the second premium at its present value, $K_1 e^{-rt_1} = 4.88$, gives what ending up with the same call costs in total: 7.90 against the vanilla's 6.81 at a 10% volatility, 22.59 against 21.79 at 50%. The total exceeds the vanilla at every volatility (committing to exercise buys the identical instrument, so it cannot be cheaper), and the gap is what the right to walk away at t_1 is worth: **1.10** at 10%, narrowing to **0.80** at 50%, because the higher the volatility the less likely the holder is to decline.

Chooser options let the holder decide at t_1 whether the option is a call or a put, both struck at K and both expiring at T. A simple chooser, one whose call and put share a strike and

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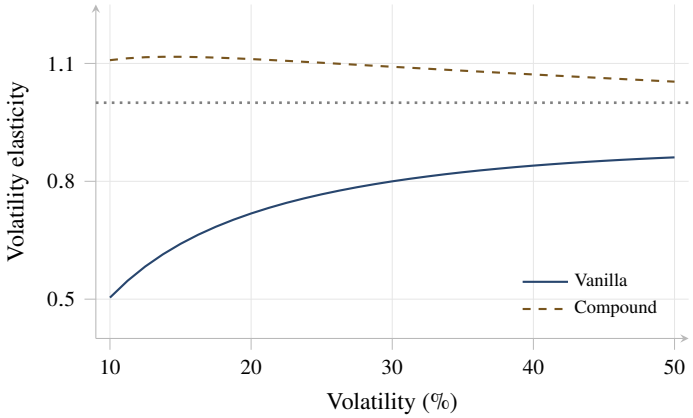


Figure 34.1: $\sigma\,(\partial V/\partial\sigma)/V$ at $S_0 = K_2 = 100$, outer premium $K_1 = 5$ paid at $t_1 = 0.5$, $T = 1$, $r = 5\%$, no dividends.

an expiration as here, decomposes exactly into vanillas. Write $c(S, K, T)$ and $p(S, K, T)$ for Black–Scholes–Merton values. At the choice date the holder takes whichever is worth more, so the chooser is then worth

$$\max\bigl\{c(S_{t_1}, K, T - t_1),\ p(S_{t_1}, K, T - t_1)\bigr\}$$

Put–call parity at t_1 gives $p - c = Ke^{-r(T-t_1)} - S_{t_1}e^{-q(T-t_1)}$, the value of a short forward, which carries no optionality at all. Substituting it,

$$\begin{aligned}\max\{c,\ p\} &= c + \max\{0,\ p - c\}\\ &= c(S_{t_1}, K, T - t_1) + e^{-q(T-t_1)}\max\Bigl(Ke^{-(r-q)(T-t_1)} - S_{t_1},\ 0\Bigr)\end{aligned}$$

where $e^{-q(T-t_1)}$ has been taken outside the maximum. Both terms are now instruments the vanilla market prices. The first is a call struck at K expiring at T, carried through the choice date untouched, and so worth $c(S_0, K, T)$ today. The second is $e^{-q(T-t_1)}$ puts struck at $Ke^{-(r-q)(T-t_1)}$, and they expire at t_1 rather than at T, because the maximum is taken on the price at the choice date. Hence

$$V_0 = c(S_0, K, T) + e^{-q(T-t_1)}\,p\Bigl(S_0,\ Ke^{-(r-q)(T-t_1)},\ t_1\Bigr) \tag{34.8}$$

Choosers are rare outright; their buyers are those who must commit to a volatility position before they know the direction.

Forward-start options are struck at the money on a future date. Under Black–Scholes their value scales with the spot, so an option struck at αS_{t_1} and expiring at T is worth

$$V_0 = e^{-qt_1}\,c(S_0,\ \alpha S_0,\ T - t_1) \tag{34.9}$$

a vanilla's price times e^{-qt_1}, with no dependence on anything observed between now and t_1. Under any realistic model the value is not this simple, because it depends on the *forward* smile—and Section 21.3 measured how badly local volatility gets that wrong. A cliquet is a strip of forward-starts, so it is the purest instrument in this chapter for trading forward volatility, and the hardest to price for exactly the same reason.

34.3 BARRIER OPTIONS

Barriers are the largest exotic market, and they take up most of this chapter.

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A barrier option is a vanilla that is extinguished (**knock-out**) or created (**knock-in**) if the underlying touches a level H. With up and down, in and out, and calls and puts, there are eight basic types, and the in-out parity

$$\text{knock-in} + \text{knock-out} = \text{vanilla} \tag{34.10}$$

means only four need to be priced.

They exist because they are cheaper. A down-and-out call costs less than the vanilla because it can be extinguished, and a buyer who believes the barrier will not be touched is paying for the part of the distribution they think matters.

All eight have closed forms under Black–Scholes with continuous monitoring, and the whole family is assembled from four expressions. Write $\phi = +1$ for a call and -1 for a put, $\eta = +1$ for a down-barrier and -1 for an up-barrier, and set

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2}, \qquad x_1 = \frac{\ln(S_0/K)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad x_2 = \frac{\ln(S_0/H)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T},$$
$$y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \qquad y_2 = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$

Then

$$\begin{aligned}
A &= \phi S_0 e^{-qT} N(\phi x_1) - \phi K e^{-rT} N\big(\phi(x_1 - \sigma\sqrt{T})\big),\\
B &= \phi S_0 e^{-qT} N(\phi x_2) - \phi K e^{-rT} N\big(\phi(x_2 - \sigma\sqrt{T})\big),\\
C &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_1) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_1 - \sigma\sqrt{T})\big),\\
D &= \phi S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(\eta y_2) - \phi K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N\big(\eta(y_2 - \sigma\sqrt{T})\big),
\end{aligned} \tag{34.11}$$

and the four knock-in values are read off the table below. Knock-outs follow from the parity (34.10).

Knock-in option	**Value, by where the barrier sits relative to the strike**			
Down-and-in call	$H \le K$:	C	$H > K$:	$A - B + D$
Up-and-in put	$H \ge K$:	C	$H < K$:	$A - B + D$
Up-and-in call	$H \ge K$:	$B - C + D$	$H < K$:	A
Down-and-in put	$H \le K$:	$B - C + D$	$H > K$:	A

Three things can be read off (34.11) directly.

A is the vanilla, since $x_1 = d_1$; the barrier enters only through B, C, and D. B is A with the barrier replacing the strike inside the normal arguments: the strike still sets the payoff, but the barrier now sets the boundary. C and D are the reflections of A and B in the barrier, weighted by the powers of H/S_0 that the reflection principle produces. That principle pairs each path that touches the barrier with its mirror image in the barrier from the moment of touching, and when the price has a drift the mirror image is not equally likely, which is what the powers correct for, and why Section 34.5's static hedge is exact only when the drift is zero.

The two entries equal to A are the vanilla itself, because a barrier lying between the spot and the strike is one the option cannot finish in the money without having crossed: an up-and-in call with $H < K$ knocks in on every path that pays it anything.

And the formula contains the static hedge. When $r = q$ the exponent λ collapses to $1/2$, the two weights become H/S_0 and S_0/H, and the down-and-in call's C reads term for term as K/H puts struck at H^2/K. Section 34.5's reflection result is visible in the formula, which is a good reason to write the formula out rather than call a library.

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The formulas assume no **rebate**, a fixed amount R paid to the holder when a knock-out option is extinguished. A rebate is valued separately, and for a down barrier both versions have closed forms in the same λ as (34.11). If it is paid at expiration it is a digital on the running minimum, paying if $m_T \le H$, and is worth Re^{-rT} times the risk-neutral probability that the barrier is touched,

$$P(m_T \le H) = N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} - (\lambda-1)\sigma\sqrt{T}\right) + \left(\frac{H}{S_0}\right)^{2\lambda-2} N\left(\frac{\ln(H/S_0)}{\sigma\sqrt{T}} + (\lambda-1)\sigma\sqrt{T}\right)$$

whose weight $(H/S_0)^{2\lambda-2}$ is the one that already appears in C and D. If it is paid on the touch it is a first-passage claim, discounted from a random date, and is worth

$$R\left[\left(\frac{H}{S_0}\right)^{\lambda-1+\psi} N(z) + \left(\frac{H}{S_0}\right)^{\lambda-1-\psi} N\big(z - 2\psi\sigma\sqrt{T}\big)\right]$$

where

$$\psi = \sqrt{(\lambda-1)^2 + \frac{2r}{\sigma^2}}, \qquad z = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \psi\sigma\sqrt{T}$$

An up barrier reverses the signs of the arguments of N in both. For the down-and-out call of Example 34.1 below, $\lambda = 1.3$, and with no dividends $\psi = \lambda$ exactly; the barrier is touched with probability 0.652, so a rebate of 1 paid at expiration is worth $e^{-0.05} \times 0.652 = 0.620$, and the same rebate paid on the touch, which arrives earlier on average, is worth 0.643. Either is added to the no-rebate value to price the contract with it.

Example 34.1 What the barrier removes

$S_0 = 100$, $K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, continuously monitored.

Vanilla call	12.335999
Down-and-out call	9.111221
Down-and-in call (the difference)	3.224778

The down-and-in value is the first row of the knock-in table, the single expression C with $\lambda = 1.3$: $(H/S_0)^{2\lambda} = 0.760380$ and $(H/S_0)^{2\lambda-2} = 0.938740$, $y_1 = -0.517884$, and

$$C = 100 \times 0.760380 N(-0.517884) - 100e^{-0.05} \times 0.938740 N(-0.767884) = 3.224778$$

The barrier removes 26.1% of the premium—a discount large enough to be the entire reason the instrument is bought.

Two observations shape everything after this.

The knock-out is not simply a cheaper call. Approaching the barrier makes the option *less* valuable while approaching the strike makes it more valuable, and between those two pulls a knock-out's delta can exceed one in magnitude and its gamma can change sign. Figure 34.4 measures both. *A reverse knock-out* (barrier above the strike, so the option is knocked out when it is deep in the money) has a discontinuous payoff at the barrier and is the most difficult vanilla-like instrument in the market.

And the price depends on the whole path distribution. The vanilla depends only on the terminal distribution, which Section 19.2 showed the option chain reveals. The barrier depends on the probability of touching, which it does not.

34.4 WHY BARRIERS BREAK TREES

Section 20.4 promised this, and the failure is instructive rather than merely inconvenient.

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Example 34.2 More steps, worse answer

Price Example 34.1's down-and-out call ($S_0 = K = 100$, $H = 90$, $r = 5\%$, $\sigma = 25\%$, $T = 1$, no dividends, European exercise) on a Cox–Ross–Rubinstein (CRR) tree, knocking out at any node at or below H. The tree can knock out only at a time step, while the analytic answer is the continuously monitored one: **9.111221**.

Steps	Tree	Error	Effective barrier
50	9.142046	+0.031	89.94
51	10.494157	**+1.383**	86.93
100	9.944477	+0.833	88.25
200	9.144117	+0.033	89.94
400	9.438414	+0.327	89.36
1600	9.153506	+0.042	89.92

Fifty-one steps is forty-five times worse than fifty. And 1600 steps is worse than 200.

The mechanism is entirely in the last column. The tree can only knock out at a node, so the barrier it actually enforces is the highest node level at or below H. As n changes, that level jumps around: at $n = 50$ it is 89.94 and at $n = 51$ it is 86.93. The tree is pricing a different contract at every step count, and the error is dominated by which contract rather than by discretization.

The error is also one-signed. The effective barrier is always at or below the true one, so the option is always over-priced, which is worse than a random error, because averaging two grids does not help (the binomial error of a vanilla alternates in sign with the parity of n, which is why averaging neighboring grids works there; here it never changes sign) and Richardson extrapolation, which Section 20.3 already warned about, makes it worse.

Now fix it. Choose n so that a layer of nodes falls exactly on the barrier. In a CRR tree the node levels are $S_0 u^{\pm j}$ with $u = e^{\sigma\sqrt{T/n}}$, so this requires

$$\frac{\ln(S_0/H)}{\sigma\sqrt{T/n}} = m \quad \text{for an integer } m \qquad \Longrightarrow \qquad n = \left\lfloor \frac{m^2\sigma^2 T}{\ln^2(S_0/H)} \right\rfloor$$

m	Steps	Tree	Error
4	90	9.111965	+0.00074
6	202	9.119059	+0.00784
8	360	9.113142	+0.00192
9	456	9.111153	**−0.00007**
12	810	9.113370	+0.00215
16	1441	9.112680	+0.00146

At 90 steps with the layer on the barrier, the error is 0.0007. At 100 steps without it, the error is 0.833—more than a thousand times larger, with more steps.

The lesson generalizes well beyond barriers. A numerical method that resolves the payoff's discontinuities converges; one that does not, does not—and refining a grid that misses the discontinuity refines the wrong problem. Section 20.3's control variate worked for the same underlying reason, and Section 20.4's trinomial tree exists to give the free parameter that makes this placement possible for any n.

Never let a user choose the step count on a barrier product. The pricing library should choose it from the barrier.

In Figure 34.2's left panel, the sawtooth is not noise: each descending run is the effective barrier drifting up toward H as n increases, and each jump is a node crossing the barrier and resetting it.

Or choose a method whose grid is free. The binomial tree's trouble is that it has no spare parameter: $u = e^{\sigma\sqrt{\Delta t}}$ is fixed once σ and Δt are, so the node levels are determined and the

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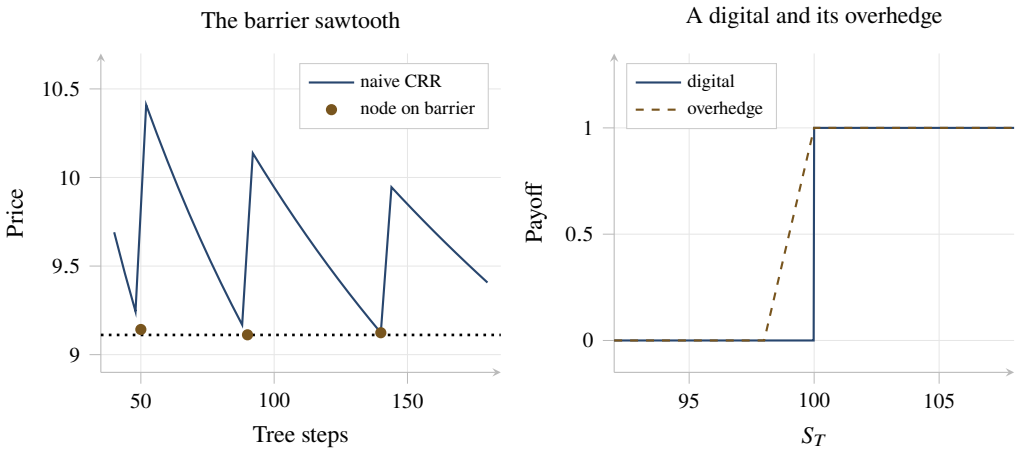


Figure 34.2: Left: Example 34.2; the dotted line is the analytic value. Right: a one-sided call spread dominating a digital.

barrier falls where it falls. A trinomial tree carries a space step that can be set independently of the time step, so choosing $\Delta x = \ln(S_0/H)/m$ for an integer m puts a layer of nodes on the barrier at *every* step count rather than at lucky ones. In $\ln S$ that tree *is* the explicit finite-difference scheme of Section 20.12, and an explicit scheme pays for the freedom with a stability condition, $\Delta x \geq \sigma\sqrt{\Delta t}$: at 50 time steps it allows only two space steps between the spot and the barrier, so hitting the barrier costs a coarse grid. An implicit scheme has no such condition. There the barrier is made the lower boundary of the grid, where the knock-out is the exact condition $V = 0$ rather than a test applied at interior nodes, and the space step is refined as far as wanted independently of the time step. That is the sense in which these are the natural methods for barriers: the discontinuity is put on the mesh by construction instead of being hunted for. On Example 34.2's contract the largest error over 50, 51, 100, 200 and 400 steps falls from **1.383** to **0.017**, and the jump from 50 steps to 51 that cost 1.35 becomes 0.02. This is why barrier books are priced on grids rather than on binomial trees.

34.5 STATIC REPLICATION

Section 18.13 argued that where gamma diverges, a static hedge is the only workable answer. Barriers are the main place that argument pays off.

Example 34.3 A barrier hedged with vanillas and never touched again

When the drift is zero ($r = q$), a **reflection** argument gives an exact static replication. A down-and-in call with strike K and barrier $H < K$ has the same value as K/H puts struck at H^2/K.

The argument has two cases. If the barrier is never touched, the call never comes alive, and the puts expire worthless too: they are struck at H^2/K, below H, and the price has stayed above H. If the barrier is touched, then at that moment K/H puts struck at H^2/K are worth exactly one call struck at K (a symmetry of the lognormal distribution that holds when $r = q$), so the puts are sold and the call bought at no cost, and the hedge has become the option.

Nothing has to be computed to confirm it, because (34.11) contains it. A down-and-in call with $H \leq K$ is the single block C, and at $r = q$ the exponent is $\lambda = 1/2$, so the two weights collapse to H/S_0 and S_0/H:

$$C = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right], \qquad y_1 = \frac{\ln\big(H^2/(S_0K)\big)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}$$

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Now price the puts. One struck at H^2/K has $d_1 = \ln\big(S_0K/H^2\big)/(\sigma\sqrt{T}) + \sigma\sqrt{T}/2$, so $-d_1 = y_1 - \sigma\sqrt{T}$ and $-d_2 = y_1$, and K/H of them are worth

$$\frac{K}{H}\left[\frac{H^2}{K}e^{-rT}N(y_1) - S_0e^{-rT}N\big(y_1 - \sigma\sqrt{T}\big)\right] = e^{-rT}\left[H\,N(y_1) - K\frac{S_0}{H}\,N\big(y_1 - \sigma\sqrt{T}\big)\right]$$

term for term the same expression. The replication is therefore exact rather than close, at any common level of $r = q$: the bracket contains no r, so raising the carry only discounts the same number. The condition is zero *drift*, not zero rates.

Now break the condition. At $r = 5\%$, $q = 0$:

Down-and-in call	3.224778
Replicating puts	1.861532
Static hedge error	**−42.3%**

The replication is not robust to drift, and 42% is not a rounding error. Refinements exist (replicating with a strip of vanillas at several maturities rather than one, which trades exactness for robustness), and they are what desks actually use.

But the reason to want a static hedge survives the error. A static hedge is a portfolio of vanillas set up once and traded again at most once, when the barrier is touched. It requires *no rebalancing*, so it has no running transaction costs and no gamma at the barrier; and it does not depend on the surface dynamics that Section 19.12 said barrier prices depend on. A hedge that is wrong by a known amount and never has to be adjusted can be better than one that is right in a model and must be adjusted precisely where the model is least reliable.

34.6 DISCRETE MONITORING

Every formula so far assumes the barrier is monitored continuously. *No contract is*: barriers are observed at a daily close, sometimes weekly or monthly. The gap between the two is larger than it looks and has a clean correction.

Example 34.4 What a daily close is worth

Broadie, Glasserman, and Kou showed that a barrier monitored m times behaves, to first order, like a *continuous* barrier shifted *away* from the spot, and pricing with the shifted barrier is the **continuity correction**:

$$H_{\text{eff}} = H\,e^{\pm\beta\sigma\sqrt{T/m}}, \qquad \beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} = 0.5826 \tag{34.12}$$

where ζ is the Riemann zeta function, $\zeta(1/2) = -1.4604$, and the sign in the exponent is chosen so the barrier moves further from the current spot: down for a down-barrier, up for an up-barrier.

Applied to Example 34.1's down-and-out call, whose continuous value is 9.111221:

Monitoring	m	**Effective** H	**Price**	**vs. continuous**
Daily	252	89.18	9.527218	+4.57%
Weekly	52	88.20	9.968231	+9.41%
Monthly	12	86.29	10.675121	+17.16%
Quarterly	4	83.68	11.365166	+24.74%

Even daily monitoring is worth 4.6% of the option, and monthly monitoring 17%. A knock-out priced as continuous is undervalued, because a path that dips below the barrier between closes is not caught.

Verify it independently. Simulating the daily-monitored contract directly with 400,000 antithetic paths:

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	Value	Against Monte Carlo
Monte Carlo (daily)	9.523044	(s.e. 0.028437)
Corrected formula (34.12)	9.527218	+0.15 standard errors
Uncorrected (continuous)	9.111221	**−14.5 standard errors**

The correction lands within a fifth of a standard error and the uncorrected formula is fourteen and a half away. The correction looks like a fudge and is not: β comes from the expected overshoot of a random walk past a level, and the same constant appears wherever a discretely observed process is compared with a continuous one.

Three consequences for practice follow.

The correction is not optional at low monitoring frequencies. Quarterly observation moves the price by a quarter, which no bid–ask absorbs.

It interacts with Section 34.4. A tree with a node layer on the *stated* barrier prices the continuous contract; to price the discrete one, the layer goes on H_{eff}. A misaligned layer and the correction both move the barrier away from spot, so an unaligned tree that skips the correction can land near the discrete price by accident: on this contract it is 0.42 high at 100 steps and 0.39 low at 50. Neither adjustment substitutes for the other, and a tree that is right for the wrong reason at one step count is wrong at the next.

And it matters most for the products of Chapter 35. An autocallable (a note that redeems early if the underlying is above a call barrier on an observation date, and exposes the investor to the underlying's loss if a lower knock-in barrier is breached) observes its knock-in barrier daily and its call barrier monthly, so a single instrument carries two different monitoring frequencies and needs two different corrections.

34.7 BARRIERS AND THE VOLATILITY SURFACE

A barrier price depends on the volatility surface in a way a vanilla price does not, and Section 21.11 measured by how much. Two models, local volatility and a stochastic-volatility model, were calibrated to the *same* vanilla surface and verified to reproduce it. They then priced a one-year up-and-out call, spot and strike at 100, barrier at 110, monitored daily: 0.8402 under the stochastic-volatility model against 0.4244 under local volatility, a difference of **49.5%** on an option worth less than a point.

The design of that experiment is Hull and Suo's (2002), set out in Section 21.11. What the spread measures is not an error in either model—both are right about every quoted vanilla—but the size of the question the market has left unanswered.

The reason is Section 19.12's. A vanilla depends on the terminal distribution and a barrier depends on the path, and two models can agree on every terminal distribution while disagreeing about how the underlying gets there. Calibration to vanillas is therefore necessary and not sufficient, and a desk that reports a barrier price to four decimals from a calibrated model is reporting the model.

Three practical responses follow, in ascending order of what they give up.

Price with the model, reserve against the range. Compute under two or three models and hold the spread as a reserve. Section 21.11's argument for reporting model risk applies exactly.

Hedge statically where possible, which converts the model question into a residual and a known error, as Section 34.5 described.

Or refuse the trade. A reverse knock-out at a barrier where the two models differ by half the premium is a trade whose price the desk does not know, and there are market conditions in which the correct response is to say so.

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34.8 DIGITALS AND THE OVERHEDGE

This section treats digitals at more length than their own market would justify, because a digital is the simplest payoff with a jump in it. Everything that makes a discontinuous payoff hard to hedge dynamically appears here with nothing else in the way: a delta that grows without bound as expiration approaches, a hedge that has to be overhedged rather than replicated, and a bid–ask spread set by what the overhedge costs. A barrier option carries the same jump at its barrier, which is why Example 34.6 returns to the argument for the reverse knock-out.

Section 12.8 showed a digital has no finite replicating portfolio. Here is what that costs.

The closed forms are the two halves of Black–Scholes taken apart. A **cash-or-nothing** call pays Q if $S_T > K$; an **asset-or-nothing** call pays S_T in the same event. Their values are

$$V_{\text{cash}} = Qe^{-rT}N(d_2), \qquad V_{\text{asset}} = S_0e^{-qT}N(d_1) \tag{34.13}$$

with d_1 and d_2 the usual Black–Scholes arguments, and the put versions replacing $N(\cdot)$ with $N(-\cdot)$. With $Q = 1$ the vanilla call is $V_{\text{asset}} - K\,V_{\text{cash}}$, which is Section 15.4's reading of $N(d_1)$ and $N(d_2)$ restated as two instruments rather than two integrals.

The formula is also the limit of the call spreads the example below prices, writing $c(K)$ for the value of a call struck at K:

$$\frac{c(K-\varepsilon/2) - c(K+\varepsilon/2)}{\varepsilon} \xrightarrow[\varepsilon\to 0]{} -\frac{\partial c}{\partial K} = e^{-rT}N(d_2) \tag{34.14}$$

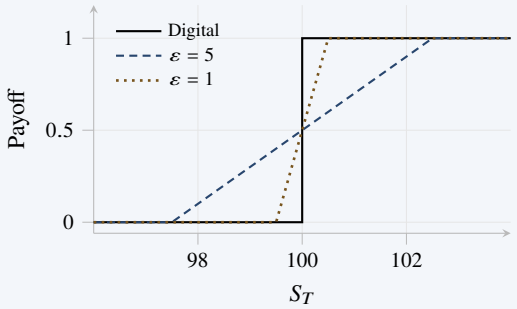
A digital is the strike-derivative of the vanilla, one differentiation short of Section 19.2's implied density, which is the second. The price is therefore a derivative of quoted prices, and a derivative of a quoted price is not itself quoted.

Example 34.5 What certainty costs

A cash-or-nothing call paying \$1 if $S_T > 100$, with $S_0 = 100$, $r = 5\%$, $\sigma = 25\%$, $T = 1$ and no dividends: worth **0.504049**.

Replicate with a call spread of width ε straddling the strike (long $K - \varepsilon/2$, short $K + \varepsilon/2$), held in quantity $1/\varepsilon$. Each spread pays at most ε, so $1/\varepsilon$ of them pay nothing below $K - \varepsilon/2$, the digital's full dollar above $K + \varepsilon/2$, and rise linearly between them. That is (34.14) evaluated at finite ε:

ε	Spreads	Value
5.00	0.2	0.504159
1.00	1.0	0.504054
0.10	10.0	0.504050
0.01	100.0	0.504049



The value converges and the position diverges. The plot shows why the value converges (the payoff approaches the step) and also why this spread is not yet a hedge: it crosses the digital, paying more below the strike and less above it.

What the dealer has sold has no finite greeks. With τ the time remaining to expiration, differentiating $e^{-r\tau}N(d_2)$ once and twice, and collecting $d_1 = d_2 + \sigma\sqrt{\tau}$,

$$\Delta = \frac{e^{-r\tau}N'(d_2)}{S\sigma\sqrt{\tau}}, \qquad \Gamma = -\frac{e^{-r\tau}N'(d_2)\,d_1}{S^2\sigma^2\tau}$$

At the money $d_1 = 1.3\,\sigma\sqrt{\tau}$, since $(r - q + \sigma^2/2)/\sigma^2 = 1.3$ here, so both greeks grow like $1/\sqrt{\tau}$: each doubles whenever the time remaining is quartered. Away from the money it is the d_1

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in the numerator that carries the gamma's sign, so the gamma changes sign at the delta's peak, $Ke^{-(r-q+\sigma^2/2)\tau}$. That point closes on the strike as expiration approaches (three cents below it with a day to run, under a hundredth of a cent with a minute), while $|\Gamma|$ five cents to either side of it reaches 32. The delta is 0.25 with a day to run, 0.65 with an hour, 5.0 with a minute and 38.8 with a second.

Holding $1/\varepsilon$ spreads caps the delta at $1/\varepsilon$, which is the whole reason for holding spreads rather than the digital; and even at that cap, with $\varepsilon = 0.10$, ten shares of stock have to be traded for every dollar of payoff while the price sits inside the window in the last moments before expiration, with no time to trade them. That is what Section 18.13's "gamma diverges" means when someone has to do it.

So dealers overhedge. To **overhedge** is to hedge with a position whose payoff is at least the sold option's everywhere, paying a known premium in place of an unhedgeable jump. Sell the digital and buy $1/\varepsilon$ call spreads struck at $K - \varepsilon$ and K, so that the upper strike stays at the digital's strike and the whole window lies *below* it. That position already pays the full dollar at K and something below it, so it dominates the digital's payoff everywhere. The table widens the window and compares each cost with the digital's 0.504049:

ε	**Strikes**	**Cost**	**Premium over the digital**
1	[99, 100]	0.511635	+1.5%
2	[98, 100]	0.519249	+3.0%
5	[95, 100]	0.542210	+7.6%

That premium is what certainty costs. A dealer who will not carry the residual cannot sell the digital below the dominating spread, and buying it cannot pay above the mirror image: a spread struck entirely *above* the strike, which the digital dominates. At a one-point window those bounds are 0.4965 and 0.5116 around a digital worth 0.5040, so no two-way price narrower than **3.0%** is hedged on both sides. What a desk actually quotes is wider: under a skew the digital's own value moves (Example 19.3 differentiates along the smile), a jump through the strike can leave the overhedge short, every one of the $1/\varepsilon$ spreads is bought at the offer, and the overhedged book ties up capital. None of these is in the model here; this is the floor the payoff alone puts under the quote. Figure 34.2's right panel shows the dominance.

The width is a business decision, not a modeling one. A tighter spread is a better price, a worse hedge, and a higher cap on the position the desk may have to trade: one share at a point wide, ten at a tenth, a hundred at a hundredth. The dealer chooses where on that trade-off to sit based on how much of the position they expect to hold to expiration.

And this generalizes to every discontinuous payoff in the chapter. A reverse knock-out is a digital in disguise. It pays a positive amount right up to the barrier and nothing beyond it, which is why Section 34.3 called it the most difficult vanilla-like instrument in the market. The overhedge for it is a barrier struck slightly further away, and the premium is much larger, as Example 34.6 shows.

Who trades digitals. Very little of the market trades a digital on its own. Most digital risk sits inside other products as a coupon trigger: each observation date on an autocallable pays its coupon only if the underlying is above a level, and a range accrual pays a day's interest only if a rate stays inside a band, so both are strips of digitals, which Chapter 35's investors are short and the issuing dealers long. The one active standalone market is in foreign exchange, where European digitals and their path-dependent relatives (the **one-touch**, which pays if the rate trades at a level at any time before expiration, and the **no-touch**, which pays if it never does) are standard dealer products. Macro hedge funds buy them to take a view on a level for a fraction of a vanilla's premium.

Professionals have little need for a listed digital, because a tight call spread already is one. Cboe listed binary options, as digitals are also called, on the S&P 500 and the VIX in 2008, and they drew almost no volume. Binaries found their buyers among retail investors instead,

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sold by offshore platforms that were the counterparty to every trade, usually at maturities of minutes or hours. The European Securities and Markets Authority restricted their sale to retail clients from July 2018 (a measure renewed in three-month steps and later carried into national rules), and Israel closed its binary-options industry outright.

The growth since 2024 has come from **event contracts**: cash-or-nothing contracts on an inflation release, a central-bank decision or an election, listed on exchanges regulated by the Commodity Futures Trading Commission and offered on prediction markets, a venue structure that is still changing as of this writing. They are digitals in the strict sense of this section, but on outcomes that no vanilla option spans. There is no call spread to overhedge them with, so Example 34.5's argument does not reach them: the price is a probability set by the order book, and the market maker carries the jump at the event rather than hedging it.

Example 34.6 The instrument that is short its own underlying

An up-and-out call struck at 100 with the barrier at 120, one year, 25% volatility, and $r = q = 0$: zero carry, so the reflection results of Section 34.5 apply exactly. Its value is the vanilla A less the up-and-in call $B - C + D$ from the third row of Section 34.3's table, by in–out parity. The vanilla is worth 9.9476; the knock-out is worth **0.6709**, *6.7% of it*. Almost the whole premium has been given away, because the option is extinguished in precisely the states where a call is valuable.

Its delta is the striking part:

Spot	Value	Delta
100	0.6709	−0.0149
110	0.4081	−0.0359
115	0.2132	−0.0414
118	0.0863	−0.0430

The delta is negative from a spot of 94.3 upward. This is a call option whose holder loses money when the underlying rises, because approaching 120 removes more value through the knock-out than it adds through moneyness. Its holder hedges by buying stock, as though short a call—the reverse of every intuition built up in Chapter 18—and the sign flips back the moment the barrier is removed.

The overhedge is where the difficulty becomes a price. Dominating the payoff means pricing and hedging the contract as a knock-out with the barrier further away, and the cost rises steeply:

Barrier used	Value	Cost over the contract
122	0.8947	+33.4%
124	1.1513	+71.6%
126	1.4377	+114.3%

Moving the barrier out by two points, 1.7% of its level, costs a third of the option's value, against 7.6% for the digital's much wider five-point window in Example 34.5. That ratio is the whole of why reverse knock-outs quote in percentage points while down-and-outs quote in basis points, and it is a property of the payoff rather than of any model: the cliff is 20 points high on an option worth 0.67.

Figure 34.3 plots it. The value is a hump whose peak sits at 94.3, six points *below* the strike, because above that level a rise in spot removes more value by raising the probability of a knock-out than it adds by moving the call further into the money, and the delta in the right panel is simply the slope of the left one, crossing zero exactly at the peak. A call option with an interior maximum is not a call option in any sense Chapter 18 would recognize, and the hedging difficulty is that the maximum is inside the trading range rather than at the edge of it.

Figure 34.4 takes the same option with a few weeks to run rather than a year, and adds the gamma. The delta climbs to 0.67 by a spot of 104, turns down, crosses −1 at 115.7, and

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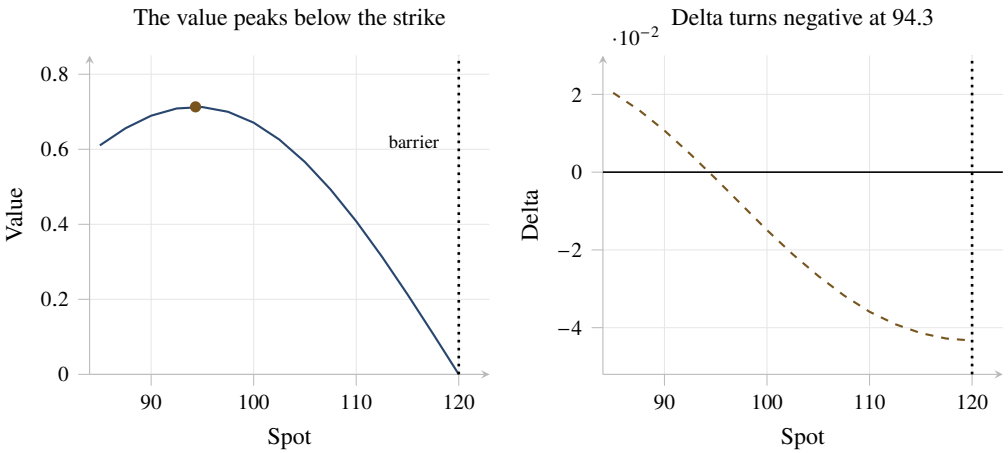


Figure 34.3: Example 34.6's up-and-out call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 1$, $r = q = 0$.

reaches -1.46 at the barrier, where a vanilla's delta is $+1$; with a year to run it never left $[-0.043, 0.027]$. The gamma is negative above 104.4, where a vanilla's is positive everywhere, and at -0.23 near a spot of 113 its magnitude is three times the largest gamma the vanilla ever has.

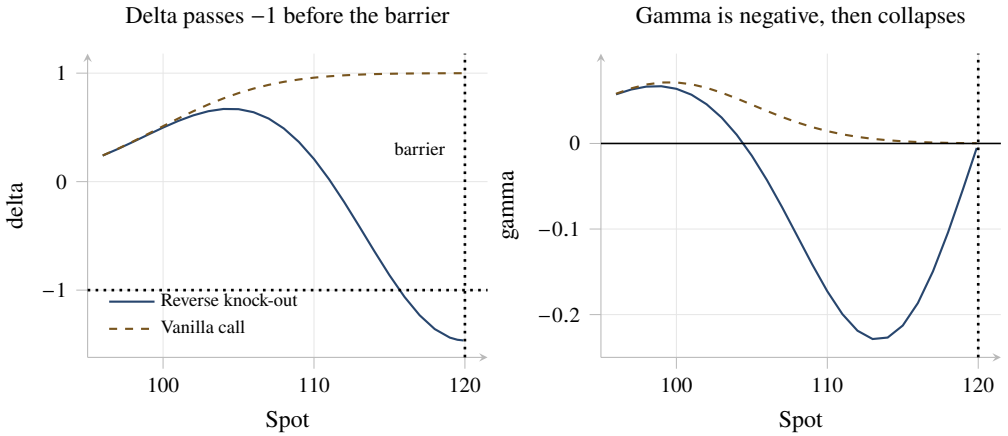


Figure 34.4: Up-and-out call against a vanilla call: $K = 100$, $H = 120$, $\sigma = 25\%$, $T = 0.05$, $r = q = 0$. The dotted vertical line is the barrier, the dotted horizontal one $\Delta = -1$.

The hedging problem is in the last few points of the right-hand panel. Approaching the barrier the value becomes nearly linear in the spot (0.146 at a spot of 119.9, against 0.146 for the straight line through the barrier), so the gamma collapses to -0.005 while the delta does not: it is still -1.46. A hedge of 1.46 shares against an option worth 0.15 means a tenth of a point of spot moves the hedge by the option's entire value, and at the barrier the option ceases to exist, so that delta has to come off at once. Gamma, which normally warns a hedger that rebalancing is about to become expensive, is closest to zero exactly where the position is most dangerous. In the terms of Chapter 18's expansion, equation (18.1), the gamma term is what announces that the delta term's coefficient is about to change, and here it is silent because the change is not gradual: at the barrier the delta jumps from -1.46 to zero, which no second-order

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term of an expansion around a smooth value can represent.

34.9 LOOKBACKS AND ASIANS

This section compares two path-dependent payoffs, one cheaper than the vanilla and one dearer. Both have closed forms, which come first.

Two averages are in use. The traded contract takes the arithmetic average of its fixings, $\frac{1}{n}\sum_{i=1}^{n} S_{t_i}$, the discrete form of the A_T in (34.3). The **geometric average** multiplies the fixings and takes the nth root,

$$G_T = \Big(\prod_{i=1}^{n} S_{t_i}\Big)^{1/n} = \exp\Big(\frac{1}{n}\sum_{i=1}^{n} \ln S_{t_i}\Big),$$

which becomes $G_T = \exp\left(\frac{1}{T}\int_0^T \ln S_t\, dt\right)$ when the averaging is continuous, and which never exceeds the arithmetic average of the same prices. Its logarithm is an average of normal variables and is therefore normal, so the geometric-average Asian call, paying $(G_T - K)^+$, is exactly lognormal: it is Black–Scholes with two substitutions. Averaging continuously over $[0, T]$,

$$c_G = S_0 e^{(b_G - r)T} N(d_1) - K e^{-rT} N(d_2), \qquad \sigma_G = \frac{\sigma}{\sqrt{3}}, \qquad b_G = \frac{1}{2}\left(r - q - \frac{\sigma^2}{6}\right) \tag{34.15}$$

with $d_1 = \left[\ln(S_0/K) + (b_G + \sigma_G^2/2)T\right]/(\sigma_G\sqrt{T})$ and $d_2 = d_1 - \sigma_G\sqrt{T}$. The $1/\sqrt{3}$ is Section 33.8's variance ratio; b_G is the drift the geometric average inherits, which is below $r - q$ because a geometric mean sits below an arithmetic one.

The arithmetic-average Asian, the one actually traded, has no closed form, because a sum of lognormals is not lognormal. Two devices cover the gap. Matching the average's first two moments to a lognormal and pricing with Black's formula gives 6.883 for the contract in Example 34.7, against 6.85 by simulation: about half a percent high, which is the size of error the whole approximation literature is arguing about, though it grows with volatility, to 1.4% at 50%. The arithmetic version is worth 5% more than the geometric one priced in that example, the gap between an arithmetic and a geometric mean, and that gap rises roughly in proportion to volatility, from 2% at a 10% volatility to 11% at 50%. And (34.15) serves as a control variate for simulation: the two payoffs correlate at 0.9994, and subtracting the geometric payoff one for one cuts the variance by a factor of 330, or about 850 with the regression-optimal coefficient. The instrument that is traded is the one without a formula, and it barely matters—which is the reverse of the situation for every barrier in this chapter.

The floating-strike lookback call paying $S_T - S_{\min}$, where $S_{\min}$ is the running minimum observed so far, is worth

$$c_{\text{lb}} = S_0 e^{-qT} N(a_1) - S_0 e^{-qT}\,\kappa\, N(-a_1) - S_{\min} e^{-rT}\left[N(a_2) - \kappa\, e^{Y_1} N(-a_3)\right], \qquad \kappa = \frac{\sigma^2}{2(r-q)} \tag{34.16}$$

where

$$a_1 = \frac{\ln(S_0/S_{\min}) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad a_3 = \frac{\ln(S_0/S_{\min}) + (-r + q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

$a_2 = a_1 - \sigma\sqrt{T}$, and $Y_1 = -2(r - q - \sigma^2/2)\ln(S_0/S_{\min})/\sigma^2$. The floating-strike put, paying $S_{\max} - S_T$ with $S_{\max}$ the running maximum, is the mirror image: the maximum takes the place of the minimum and the drift terms change sign. The formula is singular at $r = q$ (κ diverges), though the value is not, the limit being finite; an implementation that does not special-case zero drift will return nonsense on precisely the parameter set Section 34.5 cares about.

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Example 34.7 Averaging destroys optionality; looking back manufactures it

All at $S_0 = K = 100$, $\sigma = 25\%$, $r = 5\%$, $T = 1$:

	Value	Relative to vanilla
Vanilla at-the-money call	12.3360	1.00
Floating-strike lookback call	20.5522	**1.67**
Geometric-average Asian call	6.5298	**0.53**

Same underlying, same volatility, and a factor of three between the two exotics.

The Asian is cheap because an average has less variance than an endpoint. Section 33.8 computed the ratio: $(n+1)(2n+1)/6n^2$, tending to $1/3$, so the effective volatility is about $1/\sqrt{3}$ of the true one. An Asian option is a vanilla on a less volatile quantity, and that explains most of the discount: at $25\%/\sqrt{3} = 14.4\%$ an at-the-money call is worth 8.38, about two-thirds of the vanilla. The rest comes from the average's lower forward, the drift b_G of (34.15), which takes the ratio to 0.53.

The lookback is expensive because it removes the timing decision. A floating-strike lookback call pays $S_T - \min S$, so the holder buys at the best price achieved. Nothing is left to get wrong, which is worth 67% more than the vanilla.

Figure 34.5 repeats the comparison across volatilities, and it separates what is optionality from what is not. At zero volatility the vanilla and the lookback are worth the same 4.88: the path then rises steadily at the 5% rate, its minimum is the starting price, and both pay only the drift. The Asian starts lower, at 2.41, because the average of a rising path sits below its end, and that gap is carry rather than optionality. Between 25% and 50%, each volatility point adds about 0.61 to the lookback, 0.38 to the vanilla and 0.18 to the Asian.

Three practical notes follow.

Asians dominate commercial usage and lookbacks barely exist. A corporate hedging a year of purchases has an average exposure and wants an Asian; no one has an economic exposure to the minimum of a price path. Lookbacks are taught more than they are traded.

The Asian's model risk is low and the lookback's is high. An average smooths the path, so an Asian is less sensitive to the path distribution than the vanilla is to the terminal one; a lookback is a bet on the extremes and is very sensitive to it.

And discrete monitoring matters in both, in opposite directions. A monthly-averaged Asian is worth more than a continuously-averaged one (fewer observations, more variance), while a daily-monitored lookback is worth less than a continuous one, because a discretely observed minimum is higher. Section 34.6's continuity correction is the same adjustment, and the sign follows from which way discretization moves the observed extreme.

34.10 BASKETS, RAINBOWS, AND EXCHANGE OPTIONS

Multiple underlyings are the second source of genuine difficulty, and their new input is correlation.

Example 34.8 What correlation is worth

Two assets at 100 with volatilities 25% and 30%, $T = 1$, $r = 5\%$, no dividends.

A Margrabe exchange option, the right to give up asset 2 and receive asset 1, has the closed form derived in Section 28.11 as equation (28.14):

$$V_0 = S_1 e^{-q_1 T} N(d_1) - S_2 e^{-q_2 T} N(d_2), \qquad \hat{\sigma} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{34.17}$$

with $d_1 = \left[\ln\left(S_1 e^{-q_1 T}/S_2 e^{-q_2 T}\right) + \hat{\sigma}^2 T/2\right]/(\hat{\sigma}\sqrt{T})$ and $d_2 = d_1 - \hat{\sigma}\sqrt{T}$. Correlation enters only through $\hat{\sigma}$, so the table below is a table of Black–Scholes–Merton prices at five volatilities:

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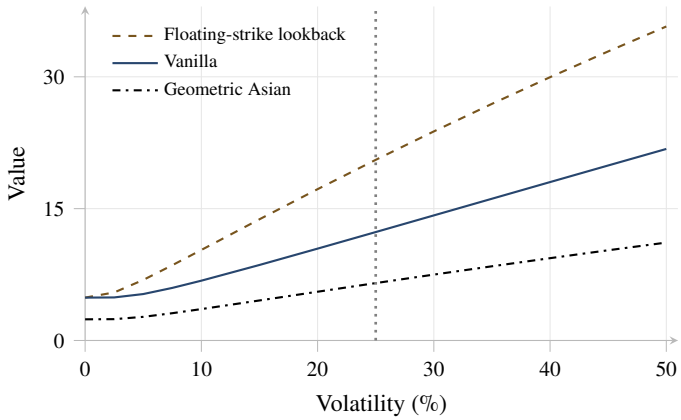


Figure 34.5: At-the-money calls against volatility: $S_0 = K = 100$, $r = 5\%$, $T = 1$, no dividends. The dotted line marks Example 34.7's 25%.

ρ	Effective volatility	Value
−0.50	47.70%	18.8495
0.00	39.05%	15.4808
+0.50	27.84%	11.0703
+0.90	13.23%	5.2737
+0.99	6.32%	2.5227

The value falls by a factor of seven across the correlation range. No interest rate appears, for the reason Section 28.11 gives: in units of asset 2 there is nothing to finance.

Now compare a *basket call* on $(S_1 + S_2)/2$ struck at 100 with a *basket of calls*, $(c_1 + c_2)/2$, half a 100-strike call on each asset:

ρ	Basket call	Basket of calls	Ratio
−0.50	8.3481	13.2836	0.628
0.00	10.3260	13.2836	0.777
+0.50	11.9088	13.2836	0.896
+0.90	13.0178	13.2836	0.980
+1.00	13.2792	13.2836	1.000

An option on a basket is always cheaper than a basket of options, with equality only at perfect correlation and equal volatilities, which is why the last row still shows a gap of 0.0044. The gap is the diversification the basket option does not let the holder keep: an option on each asset pays on whichever rises, while an option on the average pays only if the average rises.

And unlike the exchange option, the basket has no closed form, for the reason the arithmetic Asian does not: a weighted sum of lognormals is not lognormal. The workhorse approximation matches the first two moments of the basket $B = \sum_i w_i S_i$, with w_i units of asset i, to a lognormal,

$$M_1 = \sum_i w_i F_i, \qquad M_2 = \sum_i \sum_j w_i w_j F_i F_j \, e^{\rho_{ij}\sigma_i\sigma_j T}, \qquad \sigma_B = \sqrt{\frac{1}{T}\ln\frac{M_2}{M_1^2}} \tag{34.18}$$

where $F_i = S_i e^{(r-q_i)T}$, and then prices with Black's formula at forward M_1 and volatility σ_B. The approximation degrades exactly where the option is most interesting: against the exact values in the table it overstates by 0.06% at $\rho = 0.9$ and by 0.75% at $\rho = -0.5$, because a negatively correlated basket is the least lognormal. The geometric basket $\prod_i S_i^{w_i}$, by contrast, is exactly lognormal and has a closed form, which makes it the natural control variate here just as (34.15) is for the Asian.

That gap is what Section 22.11's dispersion trade, short the index option and long the

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components, trades: it earns the difference between implied and realized correlation, and the ratio column shows how much is at stake.

The modeling difficulty is that ρ has to come from somewhere. With two assets and a liquid option on each plus a liquid basket option, it can be implied. With fifty assets it cannot, because one basket price cannot determine 1,225 pairwise correlations, and the standard device (assume a single correlation for all pairs) is a one-factor approximation whose error is exactly Section 32.9's factor-reduction problem in a different market.

34.11 WORST-OF OPTIONS

One multi-asset payoff is the single most common exotic sold to retail investors, and its correlation exposure runs the opposite way to a basket's.

With two assets it has a closed form and the derivation is one identity. Since $\min(S_1, S_2) = S_1 - \max(S_1 - S_2, 0)$, a forward on the minimum is a long position in asset 1 against a short exchange option (34.17), and the strike-K put on the minimum follows from

$$p_{\min}(K) = Ke^{-rT} - \underbrace{\left[S_1 e^{-q_1 T} - V_0^{\text{exch}}\right]}_{\text{PV of receiving } \min(S_1,S_2)} + c_{\min}(K) \tag{34.19}$$

Stulz's formula supplies the last term in bivariate normal probabilities:

$$\begin{aligned} c_{\min}(K) = {} & S_1 e^{-q_1 T} M(y_1,\, -d_1;\, -\rho_1) + S_2 e^{-q_2 T} M(y_2,\, d_2;\, -\rho_2) \\ & - Ke^{-rT} M\left(y_1 - \sigma_1\sqrt{T},\; y_2 - \sigma_2\sqrt{T};\; \rho\right) \end{aligned} \tag{34.20}$$

where $\hat{\sigma}$, d_1, and d_2 are the exchange option's (34.17), and

$$y_i = \frac{\ln\left(S_i e^{-q_i T}/Ke^{-rT}\right) + \sigma_i^2 T/2}{\sigma_i\sqrt{T}}, \qquad \rho_1 = \frac{\sigma_1 - \rho\sigma_2}{\hat{\sigma}}, \qquad \rho_2 = \frac{\sigma_2 - \rho\sigma_1}{\hat{\sigma}}$$

The exchange option appears twice, once as an instrument in (34.19) and once as the correlation structure inside M, which is the sense in which every two-asset payoff in this chapter is built from Margrabe.

Three assets is where it stops. The same construction extends to n assets (Johnson's formula), but the bivariate M becomes an n-variate normal distribution function, and quadrature for those is already slow in three dimensions and unreliable above four. The three-asset worst-of that dominates the retail market is therefore simulated, which is why Example 34.9 reports a table rather than a formula, and why the note desk's risk numbers are Monte Carlo numbers with all the noise that implies.

Example 34.9 Why the coupon is high

Three assets, each at 100 with 28% volatility, $T = 1$, $r = 5\%$. A **worst-of put** struck at 100 pays on the *minimum* of the three. The benchmark is a vanilla put on any one of them, also struck at 100, which is worth 8.5955. The worst-of values are simulated, with 4 million antithetic pairs at each correlation; every standard error is below 0.005.

ρ	$E[\min S_T]$	Worst-of put	Relative to single-asset put
0.20	83.87	17.64	**2.05**
0.40	86.59	16.26	1.89
0.60	89.86	14.71	1.71
0.80	94.23	12.81	1.49
0.95	99.62	10.64	1.24

The worst-of put is worth more as correlation falls, exactly the opposite of the basket in

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Example 34.8.

A basket averages, so low correlation *reduces* dispersion and cheapens the option. A worst-of takes a minimum, so low correlation *increases* the chance that at least one asset falls a long way. At a correlation of 0.20 the expected minimum at one year is 83.87 against a starting level of 100—the drift is upward and the minimum still falls sixteen percent, purely from picking the loser three times.

A **worst-of note** is a bond whose principal is repaid in full only if the worst of the three assets finishes above the strike; otherwise the investor bears that asset's loss. The investor has therefore sold the issuer a worst-of put, and the issuer pays for it as a coupon above its own funding rate. At $\rho = 0.20$ the put is worth 17.64 per 100 of notional, roughly 18 percentage points of coupon on a one-year note, and twice what a note on any single one of the assets could pay.

So the investor in such a note is long correlation, losing as it falls: the lower the correlation, the more the put is worth and the more the investor has given away. In a crash correlation rises, which on its own would lower the put's value, but it rises because everything is falling together, and with all three assets below the strike the put is deep in the money whatever the correlation. The investor therefore carries two losses for one coupon: the one that low correlation pays for, a single asset collapsing while the others hold up, and the market-wide fall, against which the rise in correlation offers no protection.

Adding assets to a worst-of always raises the coupon and always raises the risk, and a note advertised as "linked to three blue-chip stocks" is riskier than one linked to any of them individually. Chapter 35 returns to this as the central example of the gap between how these products are described and what they are.

Figure 34.6 puts this example and Example 34.8 on one axis. Both ratios come to one at perfect correlation, where the assets move as one, and they leave it in opposite directions as correlation falls: the basket call loses value against its components, while the worst-of put gains it. A worst-of note's investor is short the upper curve, which is why the coupon is highest exactly where the assets are least alike.

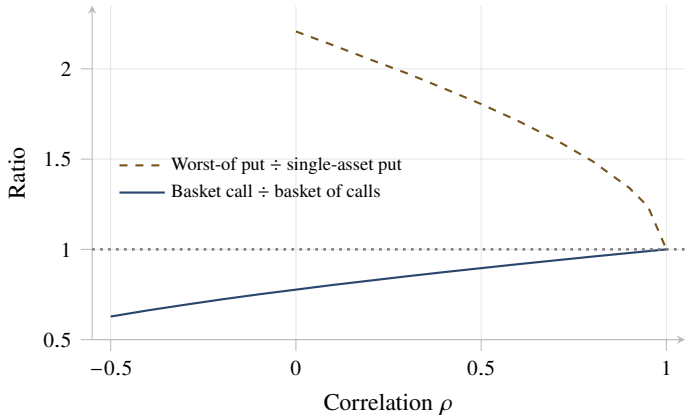


Figure 34.6: Examples 34.8 and 34.9 across correlation. The worst-of values are simulated.

34.12 CLIQUETS, AND THE CAP THAT REVERSES THE SIGN

Section 34.2 called a cliquet the purest instrument for trading forward volatility. The version actually sold is not, and the reason is instructive.

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Example 34.10 What the cap does

A three-year annual-reset cliquet paying $\sum_i \max(r_i, 0)$, where r_i is the year's return, each period capped at 8%. Under Black–Scholes each period is a forward-start at-the-money call spread, so its value depends only on that year's forward volatility. By the scaling in (34.9), the period payoff $\min\big(\max(r_i, 0),\ C\big)$ is worth, at its own reset date,

$$V = c(1,\ 1,\ \tau) - c(1,\ 1 + C,\ \tau) \tag{34.21}$$

per unit of notional, where $c(S, K, T)$ is the Black–Scholes–Merton call of Section 34.2, evaluated here at a spot of 1 and the period's forward volatility, $C = 8\%$ is the cap, and $\tau = t_i - t_{i-1}$ is the length of a period, one year here. That is a number, not a function of the level, so the whole contract is $\sum_i e^{-rt_{i-1}} V$. The table reports the undiscounted sum $3V$, which isolates the volatility comparison; discounting the second and third periods scales the total by about 0.95.

Forward volatility	Per period	Three-year cliquet
15%	0.03800	0.11401
20%	0.03670	0.11009
25%	0.03561	0.10682
30%	0.03463	0.10388
35%	0.03372	0.10115

The value falls as volatility rises. An uncapped cliquet would be long volatility; the cap makes this one short it, because the 8% cap sits close to the one-year forward and higher volatility helps the short leg of the spread more than the long one. That reversal is what lets it be sold at an attractive headline rate: it is marketed on upside participation and is, in volatility terms, a *short*-volatility position.

And the model risk is severe for a reason Section 21.3 supplies. The value sits entirely in the forward smile: the second and third periods are forward-starting, so what matters is the smile prevailing a year and two years from now. Section 21.3 measured what local volatility predicts. The skew of its local-volatility function (the slope of volatility against log-moneyness at the money) decays from -0.981 at three months to -0.231 at three years, so the model expects the smile a few years out to be far flatter than today's, while the market's forward smile is roughly stationary. A local-volatility model prices the later periods off a smile it has flattened, and a cliquet's value is concentrated exactly there.

This is the clearest case in the chapter where calibrating to today's surface is not merely insufficient but actively misleading, because the model reproduces every vanilla and gets the one quantity the product depends on structurally wrong.

Figure 34.7 shows the reversal against the contract the cap was applied to. The two slope in opposite directions, and an investor told they are buying participation in rising markets is holding the line that falls when volatility arrives. Both lines are close to straight because both options involved, the at-the-money call and the call at the cap, are struck near the forward, where an option's vega barely changes with volatility.

34.13 WHAT THEY HAVE IN COMMON

The chapter's material sorts on one question, put as a test rather than a list.

Can the payoff be replicated by traded instruments?

If it can be replicated statically (a package, a chooser, a digital as a limit of call spreads), then the vanilla market determines the price, model risk is small, and the residual is a hedging cost rather than a valuation uncertainty.

If it can only be replicated dynamically, then the price depends on the assumed dynamics, and calibration to vanillas constrains but does not determine it. Barriers, cliquets, lookbacks, and everything multi-asset live here.

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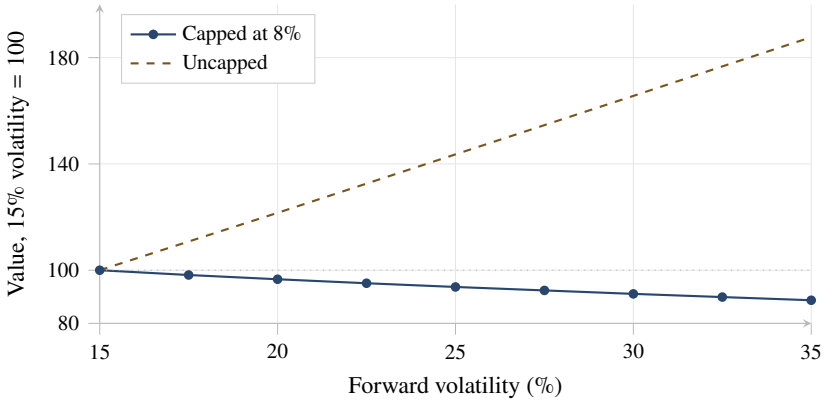


Figure 34.7: Three-year annual-reset cliquet of Example 34.10, each indexed to its own value at 15% volatility. In levels the capped contract falls from 0.11401 to 0.10115.

The test is more useful than the taxonomy of Section 34.1 because it predicts the right thing. A compound option sounds harder than a barrier and is easier; a forward-start option sounds trivial and carries as much model risk as anything in the book.

And it is a different question from whether a formula exists, which is worth putting side by side because the two are so easily confused.

Instrument	Closed form under Black–Scholes	Model risk
Package, chooser	Exact, (34.8)	None
Compound	(34.7), plus one root search	Low
Digital	(34.13)	Low; the cost is hedging
Barrier, continuous	(34.11), all eight	**Severe**
Barrier, discrete	(34.11) with (34.12)	**Severe**
Lookback	(34.16)	High
Asian, geometric	(34.15)	Low
Asian, arithmetic	**None**; moment match or simulate	Low
Exchange	(34.17)	Correlation only
Two-asset max/min	(34.20)	Correlation
Basket, worst-of ($n \geq 3$)	**None**; simulate	Correlation, unhedgeable
Forward-start, cliquet	(34.9), (34.21)	**Severe**

The two columns are almost uncorrelated. Barriers, lookbacks, and cliquets all have closed forms and carry the worst model risk in the chapter; the arithmetic Asian has no formula at all and carries almost none. A closed form is a statement about Black–Scholes, not about the market: it returns the same number every time it is called, which is precision, not accuracy. The formulas are worth having because they are fast, exact within the model, and show where the model enters, not as evidence that the price is known.

And it explains a market structure fact that otherwise looks arbitrary. The exotics with static hedges trade at narrower spreads than those without, and the ordering follows the test rather than the complexity of the payoff description.

Case Study 34.1 The structure that sold a discount and bought a multiple

The **accumulator** was sold across private banks in Hong Kong and Singapore through 2007 and into 2008, and it is this chapter's mechanism assembled into a single retail contract.

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The terms sound like a discount. The client agrees to buy a fixed quantity of a stock on every trading day for a year, at a strike set a few percent below the spot price at inception. Nothing is paid at the outset. Two clauses complete it: the contract *knocks out* if the stock trades a few percent above the initial spot, and the daily quantity *doubles* if the stock is below the strike.

Decomposed, it is three of this chapter's instruments. A strip of daily forwards at a discount; short a strip of up-and-in calls, which is what the knock-out clause amounts to, since it surrenders the forwards' gains once the barrier trades; and short a leveraged strip of puts, which is what the doubling clause is. The discount is the premium of the options the client has sold, and the client is short volatility, short the downside at twice the size, and capped on the upside at a few percent.

What happened is what the structure specifies. Hong Kong and mainland equities peaked in October 2007 and fell through 2008. Because prices fell, the knock-out never triggered, so the contracts ran their full term rather than terminating early as they did in the case the salesman described. The doubling clause then required the client to buy twice the daily quantity, every day, at a strike far above the market, with the obligation growing precisely as the losses grew. The instrument acquired a nickname that was a pun on its own name: *I kill you later*.

The failure was not mathematical. Nothing here needed a model the seller did not have. The knock-out was disclosed, the doubling was disclosed, and both behaved exactly as written. What was not conveyed is the asymmetry the two clauses create together—a payoff described by its best state and sized by its worst.

Three lessons connect it to the rest of the book. A barrier that removes the upside is worth more than it looks when it is one of two clauses rather than the only one. A zero-premium structure has a premium; Chapter 35's treatment of a note's estimated initial value gives the machinery for finding it, and here it is the discount itself. And leverage written into a contract's quantity is harder to see than leverage written into its notional, because nothing in the term sheet is labeled with a multiple.

34.14 WHO TRADES EXOTICS, AND WHY

Corporate hedgers are the economically motivated end of the market. An Asian option matches an averaged exposure and a barrier matches a hedge that is only wanted in some states: an importer who needs FX protection but not if the currency has already collapsed past the point where the business is unviable. These are exotics bought because the vanilla is the wrong shape.

Structured-note issuers are the volume. Chapter 35 is about them, and almost every payoff in this chapter appears inside a retail note: barriers as knock-in protection, digitals as coupon triggers, baskets as "worst-of" underlyings, cliquets as annual ratchets.

Dealers intermediate, and the pattern is one-directional. The client is almost always short the exotic optionality and the dealer long it, because the exotic feature is what cheapens the note or the hedge. So dealers run a structural long-exotic, short-vanilla book, which they hedge in the vanilla market, and the marks Section 19.8 traces on the surface and Section 22.11's implied correlation both reflect that flow.

Hedge funds take the other side selectively, and the trades named in this book (dispersion in Section 22.11, forward-volatility positions against cliquets) are mostly bets that the dealer's hedging flow has pushed a price away from fair.

The recurring structure is worth naming once. An exotic option transfers a risk the client does not want to price to a dealer who does, and the dealer's compensation is the difference

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between what the exotic can be hedged for and what it was sold for. Sections 34.5 and 34.13 determine how large that difference has to be, which is the practical reason the test in this chapter matters.

SUMMARY

Exotics divide by path dependence, multiple underlyings, and non-standard exercise, but the useful division asks whether the payoff can be replicated by traded instruments. Statically replicable payoffs are priced by the vanilla market with small model risk; dynamically replicable ones depend on assumed dynamics, which calibration to vanillas constrains without determining.

A down-and-out call is worth well below the vanilla, and a reverse knock-out loses value as the underlying rises toward its barrier. Barriers are hard to compute as well as to hedge: a binomial tree's value jumps erratically with the number of steps unless a node layer sits on the barrier, and correcting for daily rather than continuous monitoring moves the price materially. The reflection hedge is exact only when $r = q$ and errs badly when they differ, and two models calibrated to the same vanilla surface priced one barrier far apart.

A digital is the limit of call spreads whose value converges while the position diverges, so dealers overhedge with a spread that dominates the payoff, at a modest premium over the digital. Every discontinuous payoff in the chapter is priced this way.

A lookback is worth substantially more than the vanilla and a geometric Asian substantially less: an Asian is a vanilla on a less volatile quantity, and a lookback removes the timing decision.

Correlation enters every multi-asset payoff and is rarely quoted, so an option on a basket is cheaper than a basket of options. A worst-of put on several assets is worth far more than a single-asset put, because a minimum rewards dispersion where a basket's average punishes it, and that premium is what pays a worst-of note's coupon. A capped cliquet is short forward volatility, losing value as volatility rises, and its value sits in the forward smile, which today's surface does not pin down.

A closed form is a statement about Black–Scholes, not about the market. Barriers, lookbacks and cliquets have exact formulas and the worst model risk in the chapter; the arithmetic Asian has no formula and almost none.

Corporates buy exotics because the vanilla is the wrong shape, and structured-note issuers supply the volume. Clients are almost always short the exotic optionality, so dealers run a long-exotic, short-vanilla book hedged in the vanilla market.

FURTHER READING

Zhang's *Exotic Options* and Haug's *Complete Guide to Option Pricing Formulas* are the reference works for closed forms; Haug states each formula with a worked number, which makes it the place to check an implementation against, and both should be read knowing that a closed form exists only under assumptions the market violates. Derman, Ergener and Kani on static replication of barriers is the paper behind Section 34.5 and develops the strip-of-maturities refinement this chapter only mentions. Carr and Chou extend the reflection argument considerably further. Boyle and Lau (1994) is the source of Example 34.2's node-placement rule. Broadie, Glasserman and Kou on discrete barrier monitoring supplies the continuity correction of Section 34.6. On the multi-asset material, Margrabe's original paper is short and is the source of the exchange option; Stulz (1982) is the source of (34.20), and states put prices through the identity (34.19); and Johnson (1987) extends the result to n assets. The literature on basket approximations is large, and its methods differ mainly in accuracy, so decide what accuracy is needed before sampling it. For the model-risk question of Section 34.7, the practitioner literature on barrier reserves is thin and mostly internal; the academic treatment

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under uncertain volatility is the public substitute.

PRACTICE QUESTIONS

34.1 State the replication test that organizes this chapter, and use it to explain why a compound option carries less model risk than a forward-start.

34.2 Using Example 34.1, state the fraction of the vanilla premium the barrier removes, and explain why knock-out deltas can exceed one.

34.3 Using Example 34.2, state the tree prices at 50 and 51 steps, explain the difference, and state the general lesson in a form that does not mention barriers.

34.4 Explain why grid-averaging and Richardson extrapolation both make the barrier tree error worse rather than better.

34.5 Derive the step count that places a node layer on the barrier, and state the error it achieves against an unaligned grid with more steps.

34.6 Using Example 34.3, state the static hedge error when $r = 5\%$ and $q = 0$, and explain why the condition for exactness is zero drift rather than zero rates.

34.7 Explain why a static hedge that is wrong by a known amount can be preferable to a dynamic hedge that is right in a model.

34.8 Explain why two models calibrated to the same vanilla surface can price a barrier 49.5% apart, and state the three defensible responses.

34.9 Using Example 34.5, state the cost of the $[95, 100]$ overhedge relative to the digital, and explain why the replicating call spread's notional diverges while its value converges.

34.10 Explain why an Asian option is worth about half a vanilla and a lookback about 1.67 times it, and why Asians dominate commercial usage while lookbacks do not.

34.11 Using Example 34.4, state the value of daily and of monthly monitoring relative to continuous, and explain why an unaligned tree that omits the continuity correction can land near the right price for the wrong reason.

34.12 Using Example 34.9, state the worst-of put's value relative to a single-asset put at $\rho = 0.20$, and explain why the correlation dependence runs opposite to a basket's.

34.13 Using Example 34.8, state the basket call's value relative to the basket of calls at $\rho = -0.5$ and at $\rho = 1$, explain what the gap represents, and explain why fifty assets cannot be handled with a single correlation.

34.14 Explain why a capped cliquet is short volatility, why its value sits entirely in the forward smile, and why that makes calibration to today's surface actively misleading rather than merely insufficient.

34.15 Explain why dealers run a structural long-exotic, short-vanilla book, and why exotics with static hedges trade at narrower spreads than others.

34.16 Using (34.11), price the down-and-in call of Example 34.1 from the single expression C, and explain why the up-and-in call with $H < K$ is worth the vanilla.

34.17 Name three instruments in this chapter with closed forms and severe model risk, and one with no closed form and very little. Explain why the two properties are nearly unrelated.

FURTHER QUESTIONS

34.18 Set $r = q$ in (34.11) and show that the down-and-in call's C is exactly K/H puts struck at H^2/K. Then show where the argument fails when $r \neq q$, and relate the failure to the

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exponents 2λ and $2\lambda - 2$.

34.19 Example 34.2 shows a thousandfold error reduction from node placement. Derive the effective barrier as a function of n, show that the resulting price error is first-order in the barrier displacement, and use that to explain the sawtooth's shape between jumps.

34.20 Derive the reflection result of Example 34.3 from the symmetry of the lognormal density under $r = q$. Then determine how the static hedge error scales with $(r - q)T$, and design a multi-maturity strip that reduces it, stating what it costs.

34.21 Section 34.7 reports a 49.5% model spread on a barrier. Propose a reserving policy: specify which models define the range, how the reserve is released over the trade's life, and how you would prevent the policy from being gamed by choosing convenient models.

34.22 Example 34.5 shows the overhedge premium rising with the spread's width. Derive the dealer's optimal width as a trade-off between the premium given up and the expected hedging loss, and state how it depends on the expected holding period.

34.23 Example 34.8 shows a basket option below a basket of options at all $\rho < 1$. Prove the inequality in general, determine the conditions for equality, and connect the result to the convexity argument of Section 11.3.

34.24 This chapter argues that spreads follow replicability rather than payoff complexity. Test the claim against the exotics in this chapter: rank them by replicability, rank them by the spreads you would expect, and identify any instrument where the two orderings disagree. Then explain the disagreement.

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